Example 3.19 Find the area of square whose side length is $3m + 2n - 4l$

Solution

Area of square = side × side

$= (3m + 2n - 4l) \times (3m + 2n - 4l)$

$= (3m + 2n - 4l)^2$

substituting
$a = 3m$,
$b = 2n$
$c = -4l$

We know that, $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$

$$\left[3m + 2n + (-4l)\right]^2 = (3m)^2 + (2n)^2 + (-4l)^2 + 2(3m)(2n) + 2(2n)(-4l) + 2(-4l)(3m)$$

$$= 9m^2 + 4n^2 + 16l^2 + 12mn - 16ln - 24lm$$

Therefore, Area of square $= [9m^2 + 4n^2 + 16l^2 + 12mn - 16ln - 24lm]$ sq.units.

3.4.2 Identities involving Product of Three Binomials

$$(x + a)(x + b)(x + c) = [(x + a)\ (x + b)]\ (x + c)$$

$$= [x^2 + (a + b)x + ab](x + c)$$

$$= x^2(x) + (a + b)(x)(x) + abx + x^2c + (a + b)(x)c + abc$$

$$= x^3 + ax^2 + bx^2 + abx + cx^2 + acx + bcx + abc$$

$$= x^3 + (a + b + c)x^2 + (ab + bc + ca)x + abc$$

Thus, $(x + a)(x + b)(x + c) \equiv x^3 + (a + b + c)x^2 + (ab + bc + ca)x + abc$

Example 3.20 Expand the following:

(i) $(x + 5)(x + 6)(x + 4)$ (ii) $(3x - 1)(3x + 2)(3x - 4)$

Solution

We know that $(x + a)(x + b)(x + c) = x^3 + (a + b + c)x^2 + (ab + bc + ca)x + abc$ --(1)

(i) $(x + 5)(x + 6)(x + 4)$

$= x^3 + (5 + 6 + 4)x^2 + (30 + 24 + 20)x + (5)(6)(4)$

$= x^3 + 15x^2 + 74x + 120$

Replace
a by 5
b by 6
c by 4 in (1)

(ii) $(3x - 1)(3x + 2)(3x - 4)$

$= (3x)^3 + (-1 + 2 - 4)(3x)^2 + (-2 - 8 + 4)(3x) + (-1)(2)(-4)$

$= 27x^3 + (-3)9x^2 + (-6)(3x) + 8$

$= 27x^3 - 27x^2 - 18x + 8$

Replace
x by $3x$, a by -1,
b by 2, c by -4
in (1)

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